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Environmental Chemical Exposures in Boxer Dogs With Multicentric Lymphoma

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ABSTRACT

Background: Multicentric lymphoma (ML) in dogs has similar clinical features to human non-Hodgkin lymphoma (NHL). NHL has been linked to herbicides and volatile organic compounds (VOCs), direct exposures to these chemicals have not been evaluated in ML.

Hypothesis/Objectives: ML in pet dogs was associated with herbicide and VOC exposures.

Animals: Twenty pet boxer dogs with ML and 20 unaffected pet boxer dogs enrolled across the United States between 2021 and 2024.

Methods: Prospective case–control study. Dog owners completed an environmental questionnaire and submitted dog urine, drinking water, and household air samples for analysis of the herbicides 2,4-D, glyphosate, and atrazine and the VOCs benzene, 1,3 butadiene, and trichloroethylene. Area Deprivation Indices (ADI) and USDA Rural–Urban Continuum Codes (RUCC) were assigned to all dogs based on home address.

Results: Boxers with ML were more likely to live in more urban counties (median RUCC 1, range 1–6 for cases, vs. median 2, range 1–8 for controls; $p = 0.032$). Urinary concentrations of the benzene metabolite S-phenylmercapturic acid (PHMA) were higher in dogs with ML (median 1.03 ng/mg creat, range 0.00–1.53) compared to control dogs (0.54 ng/mg creat, range 0.09–4.73; $p = 0.031$). Odds ratios for exposure to the benzene metabolite MUCA were higher in cases vs. controls, even when controlled for urbanicity (OR 1.01, 95% CI 1.00–1.02; $p = 0.035$).

Conclusions and Clinical Importance: ML in boxers might be associated with benzene exposures. If this association is causative, indoor benzene exposures could be remediated with activated carbon air filtration and certain house plants.

Abbreviations: ADI, Area Deprivation Index; DHBM, N-acetyl-S-(3,4-dihydroxybutyl)-L-cysteine; GC/MS, gas chromatography/mass spectrometry; MHB3, N-acetyl-S-(4-hydroxy-2-buten-1-yl)-L-cysteine; ML, multicentric lymphoma; MUCA, muconic acid; NHL, non-Hodgkin lymphoma; PARR, PCR for antigen receptor rearrangements; PHMA, N-acetyl-S-(phenyl)-L-cysteine; RUCC, USDA Rural–Urban Continuum Codes; VOCs, volatile organic compounds.

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1 | Introduction

Multicentric lymphoma (ML) is a common and typically fatal cancer in pet dogs [1, 2]. Both ML and human non-Hodgkin lymphoma (NHL) share clinical features and common geographic distributions [3–5]. NHL and other human lymphoid malignancies are associated with occupational exposures to volatile organic compounds (VOCs) such as benzene, 1,3-butadiene, and trichloroethylene [6–9]. Non-occupational VOC exposures can occur from air pollution and second-hand smoke, as well as from household solvents, paints, and glues in home building materials and furniture [10]. NHL is also linked to the herbicides 2,4-D, glyphosate, and atrazine [11–15], as well as nitrate in water [16, 17].

Like NHL, ML is associated with commercially applied herbicides and pesticides [18–21], household use of paints and solvents (which contain VOCs) [22], second-hand tobacco smoke [23], and proximity to waste incinerators or polluted sites [4, 22, 23]. Specific chemicals, however, have been difficult to implicate because studies in dogs with ML have not incorporated direct biomonitoring.

Several dog breeds have a higher incidence of ML [24], with boxers at higher risk for T-cell lymphoma [4]. T-cell lymphoma has distinct geographic distributions even within a single dog breed [25], which suggests potentially modifiable environmental factors.

Our overall hypothesis is that multicentric lymphoma is associated with environmental chemical exposures in pet dogs within a single breed. The primary aim of this study was to determine whether environmental and in vivo concentrations of the VOCs benzene, 1,3-butadiene, and trichloroethylene, and the herbicides 2,4-D, glyphosate, and atrazine, along with nitrate, were higher in boxers with ML compared to unaffected control boxers. A secondary aim was to determine whether urinary chemical concentrations correlate with urbanicity, or with VOCs and herbicides detected in drinking water or indoor household air.

2 | Materials and Methods

2.1 | Case Enrollment

Boxer dogs with confirmed multicentric lymphoma were recruited from the teaching hospital at the University of Wisconsin-Madison and through outreach to veterinary oncologists across other US teaching hospitals and the national VCA Animal Hospital network. In addition, study flyer information was attached to diagnostic reports of ML in purebred boxers with the assistance of Dr. Anne Avery at the Clinical Hematopathology Laboratory at Colorado State University and Drs. Jennifer Wilcox and Jorden Manasse at Antech Diagnostics.

The diagnosis of ML was based on disease distribution and cytology or histopathology; immunophenotyping by PCR for antigen receptor rearrangements (PARR) or flow cytometry was not required but was included where available. Confirmed cases of ML had to be purebred boxers based on the medical record but could be of any age and sex/neuter status.

At the time of diagnosis, attending veterinarians were asked to provide study flyers to boxer owners, who were invited to directly contact study personnel by email. Either the veterinarian or the owner provided study personnel with diagnostic reports. Recruitment and consent for owners of qualifying dogs were obtained by email using Institutional Care and Use Committee (IACUC)-approved verbiage and consent forms. Cases were enrolled within 1 month of lymphoma diagnosis, typically prior to chemotherapy.

2.2 | Control Enrollment

Unaffected purebred boxer dogs of comparable age and sex distribution were recruited from the University of Wisconsin-Madison teaching hospital and through outreach to veterinarians, boxer owners, boxer rescue associations, and boxer breed clubs through social media. Controls could have no history of systemic cancer; boxers with a history of localized cutaneous mast cell tumors were still eligible. Controls were not fully pair-matched but were enrolled concurrently with ML cases, for comparable characteristics with cases by age, sex/neuter status, and season of recruitment (i.e., December–February, March–May, June–August, or September–November). All cases and controls were recruited between April 2021 and April 2024.

2.3 | Urine, Drinking Water, and Air Sampling

A study kit was mailed to all consenting owners; the kit contained the study questionnaire, a copy of the IACUC-approved consent form for the owner's records, and materials to collect the dog's urine and drinking water. First morning urine was recommended. Tap water was run for 2 min before collecting the sample; bottled or fridge-filtered water was collected directly. All kit and sampling materials were encoded with study numbers to protect participant privacy. Samples were returned on a cold pack by overnight mail. Upon receipt, urine samples were centrifuged and then frozen at −80°C, and water was stored at 4°C until analyses.

After urine and water samples were received by research personnel, owners were sent a 6 L passive air sampling canister (Entech Instruments Inc.) to collect indoor air for benzene and trichloroethylene measurements. Owners placed the canister on their kitchen counter, opened the sampling valve to collect air, and closed the valve after 8 h. Canisters were returned in a postage-paid box directly to the analytical laboratory.

2.4 | Chemical Analyses

The herbicides 2,4-D, glyphosate, and atrazine were measured in canine urine and in drinking water using commercial ELISA kits (Abraxis Gold Standard Diagnostics) at the Wisconsin State Laboratory of Hygiene. Nitrates/nitrites were measured in urine and drinking water using automated colorimetry [26].

Benzene and trichloroethylene in indoor air were measured on a fee-for-service basis using gas chromatography/mass spectrometry (GC/MS) at the Eastern Research Group Laboratory of the

US Environmental Protection Agency [27]. Trichloroethylene in drinking water was also measured by GC/MS [28].

Stable urinary metabolites of benzene, including PHMA (N-acetyl-S-(phenyl)-L-cysteine) and MUCA (muconic acid) were quantitated in urine by ultra-performance liquid chromatography with electrospray tandem mass spectrometry; [29] on a fee-for-service basis at the Centers for Disease Control Volatile Organic Compounds Laboratory. Stable urinary metabolites 1,3-butadiene, including MHB3 (N-acetyl-S-(4-hydroxy-2-buten-1-yl)-L-cysteine) and DHBM (N-acetyl-S-(3,4-dihydroxybutyl)-L-cysteine) were also assayed using ultra-performance liquid chromatography with electrospray tandem mass spectrometry; [30] at the same laboratory. All urinary chemicals were normalized to urine creatinine to control for urine concentration [31].

2.5 | Household Environmental Questionnaire

Owners of enrolled dogs were asked to complete a questionnaire about household exposures relevant to the chemicals of interest over the year prior to enrollment. Questionnaires included three sections: (1) dog demographic data and primary residence; (2) dog's living environment, including type of neighborhood, level of drive-by vehicle traffic, use of herbicides and other pesticides, household smoking status, and use of a fireplace or wood-burning stove; and (3) dog's health, including any current clinical signs and the dog's diet and primary source of drinking water.

2.6 | Socioeconomic Deprivation and Urbanicity

Because the incidence of NHL in people has been associated with poorer neighborhoods [32] and urbanicity [33], which in turn have been associated with higher environmental exposures to VOCs [34, 35], we assessed the socioeconomic status of neighborhoods in which enrolled dogs lived for our study. We used a publicly available database, the Neighborhood Atlas [36], which assigns an Area Deprivation Index (ADI) for households based on their home addresses. A higher ADI and larger national percentiles indicate higher poverty and deprivation based on income, education, employment, and housing quality. Compiled data from the Neighborhood Atlas were available from 2017 to 2021. To assess urbanicity, in addition to the owner questionnaire, we assigned USDA Rural–Urban Continuum Codes (RUCC) based on each dog's home address; for this code system, 1 is most urban and 9 is most rural [33]. USDA Rural–Urban Continuum Code data were available for 2023 [37].

2.7 | Statistical Analyses

Questionnaire data were encoded as categorical variables, which were compared between cases and controls using Fisher's exact tests, with generation of odds ratios (OR) and 95% confidence intervals. Urinary herbicides and VOC metabolites, drinking water chemicals, indoor air VOCs, Area Deprivation Index percentiles, and RUCC were compared between cases and controls using Mann Whitney U tests. Aggregate molar urinary herbicides were compared by owner reported herbicide use and by season of sample collection using Mann Whitney U and Kruskal Wallis tests, respectively. Correlations between RUCC and indoor air benzene, aggregate urinary benzene metabolites, and aggregate urinary 1,3-butadiene metabolites were performed using Spearman rank order correlation. All analyses were performed using commercial software (Prism, GraphPad Software LLC), with $p < 0.05$ considered significant.

Multivariable logistic regression was also used to generate odds ratios (OR) and 95% confidence intervals (CI) around the relationship between VOC exposures and ML outcomes when considering relevant covariates. We built multivariable logistic regression models using a backward stepwise approach for covariates of interest with $p < 0.20$ in univariable assessments.

2.8 | Sample Size Calculation

We used the primary outcome of urinary benzene concentrations to determine sample size for this study. We based our effect size on a study of ambient (non-occupational) benzene exposure in people that found a significant increase in NHL risk with a less than two-fold increase in airborne benzene exposure [38]. With an assumption that urinary benzene metabolites would parallel airborne benzene exposures [38], we calculated that 20 boxers with ML and 20 unaffected boxer controls would provide more than 80% power and 95% confidence to detect this difference as significant in our study group.

3 | Results

3.1 | Study Group Demographics

All enrolled dogs were purebred boxers. A total of 32 dogs with lymphoma were initially enrolled. Nine boxers were excluded for not returning testing kits, two for not having a final diagnosis of multicentric lymphoma, and one for not being purebred, leaving 20 eligible enrolled boxers with ML. Twenty-eight unaffected control boxers were initially enrolled. Five were excluded for not returning test kits, and three did not match the age characteristics of the case group, leaving 20 eligible enrolled unaffected control boxers. We did not recruit more than one dog from any given household.

The median age at enrollment for cases was 6.6 years (range 3–12 years) and 8.5 years for controls (range 3–11; $p = 0.43$; Table 1). Dogs had been owned by the current family for a median of 6 to 6.5 years and had lived in their current home for a median of 5 to 5.5 years (Table 1).

3.2 | Household Environmental Questionnaire Responses

Boxers with ML were significantly more likely to live in an urban or suburban neighborhood (90%) as reported by their owners, compared to control boxers (53%; OR 7.65, 95% CI = 1.29–38.65, $p = 0.03$; Table 2). Most owners reported that their dogs drank primarily municipal water, with no observed differences between cases and controls (Table 2). There were no statistically

TABLE 1 | Demographic data from boxer dogs with multicentric lymphoma and unaffected boxer controls, enrolled across the United States from 2021 to 2024.

Signalment	Lymphoma ($n=20$)	Controls ($n=20$)
Median age (range)	6.6 (3.8–12) years	8.5 (3–11) years
Sex		
FS	6	7
FI	1	2
MN	13	10
MI	0	1
Diagnosis		N/A
Cytology or histopathology	7	
Cytology or histopathology plus immunophenotyping with PARR or flow cytometry	12	
Other[a]	1	
Season of sample collection	Winter 7 (35%)	Winter 4 (20%)
	Spring 5 (25%)	Spring 11 (55%)
	Summer 4 (20%)	Summer 3 (15%)
	Fall 4 (20%)	Fall 2 (10%)
Median years owned (range)	6.5 years (2.7–12 years)	6 years (0.7–10 years)
Median years in current home (range)	5.5 years (0.4–10 years)	5.0 years (0.7–9.8 years)
Area Deprivation Index as national percentiles[b]	Median 28.0% (Range, 2.0%–86.0%)	Median 46.5%* (Range, 4.0%–83.0%)

Abbreviations: FS, female spayed; FI, female intact; MN, male neutered; MI, male intact.
[a]Round cell neoplasia (probable lymphoma), with cranial mediastinal mass, splenomegaly with hypoechoic infiltration, peripheral hypoechoic abdominal lymphadenopathy, and hypercalcemia.
[b]A higher percentile indicates greater poverty. *$p=0.048$.

significant differences between cases and controls in reported neighborhood traffic, household smoking, use of a fireplace or wood-burning stove, or reported herbicide or insecticide use (Table 2).

3.3 | Urine and Household Herbicide Exposures

The herbicides 2,4-D, glyphosate, and atrazine were detected in the urine of all pet boxer dogs. Urinary 2,4-D concentrations were observably but not significantly higher in dogs with ML (median 12.9 ng/mg creat, range 6.2–28.9 ng/mg creat) compared to controls (9.1 ng/mg creat, range 2.4–32.0 ng/mg creat; $p=0.07$; Figure 1A). Urinary glyphosate did not differ significantly between cases (median 7.7 ng/mg creat, range 1.0–21.2 ng/mg creat) and controls (5.6 ng/mg creat, range 1.4–20.7 ng/mg creat); $p=0.64$; (Figure 1B); similar negative results were found for urinary atrazine (cases: median 0.94 ng/mg creat, range 0.08–1.83 ng/mg creat; controls: median 0.49 ng/mg creat, range 0.06–2.05 ng/mg creat; $p=0.29$; Figure 1C).

We also evaluated aggregate urinary concentrations of all three herbicides on a molar basis. These also did not differ between cases and controls (medians for both of 0.11 nmol/mg/min; $p=0.39$). Surprisingly, aggregate urinary herbicide concentrations did not vary according to the owner's reported home use of herbicides ($p=0.46$; Figure S1A) nor the season of sample collection ($p=0.35$; Figure S1B).

None of the measured drinking water samples contained any of the three herbicides, with limits of detection of 2 ppb for 2,4-D, 0.075 ppb for glyphosate, and 0.046 ppb for atrazine. Drinking water nitrate/nitrite contents were not different between ML households (median 0.59 ppm, range 0.15–7.89 ppm) and control households (median 0.94 ppm, range 0.15–16.1 ppm; $p=0.11$).

3.4 | Urine and Household VOC Exposures

The major urinary metabolite of benzene, MUCA, was detectable in all dogs; PHMA was detectable in all but one dog (Figure 2). Urinary PHMA concentrations were nearly 2-fold higher in ML cases (median 1.03 ng/mg creat, range 0.00–1.53 ng/mg creat) compared to controls (median 0.54 ng/mg creat, range 0.09–4.73 ng/mg create; $p=0.031$; Figure 2A). Median urinary MUCA concentrations were also nearly 2-fold higher in cases (114.5 ng/mg creat, range 3.3–626.5 ng/mg creat) compared to controls (66.5 ng/mg creat, range

TABLE 2 | Selected household exposures based on environmental questionnaire responses from owners of boxer dogs with multicentric lymphoma and unaffected controls, enrolled across the United States from 2021 to 2024.

Owner reported data		Lymphoma *n* (%)	Controls *n* (%)	OR	95% CI	*p*
Neighborhood	Respondents	19	19			
	Rural/Farm	2 (10.5)	9 (47.4)			
	Urban/Suburban	17 (89.5)	10 (52.6)	7.65	1.29–38.65	**0.03**
Drive-by traffic[a]	Respondents	19	20			
	Minimal	5 (26.3)	10 (50.0)			
	Moderate/Heavy	14 (73.7)	10 (50.0)	2.80	0.74–10.67	0.19
Household smoking	Respondents	19	20			
	No	17 (89.5)	18 (90.0)			
	Yes	2 (10.5)	2 (10.0)	1.06	0.15–7.33	> 0.99
Primary drinking water source	Respondents	19	20			
	Non well (municipal, bottled or other)	16 (84.2)	11 (55.0)			
	Well	3 (15.8)	9 (45.0)	0.23	0.06–1.08	0.08
Fireplace or wood-burning stove	Respondents	19	20			
	No	15 (78.9)	13 (65.0)			
	Yes	4 (21.1)	7 (35.0)	0.50	0.14–2.00	0.48
Herbicide use in past year	Respondents	18	20			
	No	9 (50.0)	13 (65.0)			
	Yes	9 (50.0)	7 (35.0)	1.86	0.51–6.30	0.51
Insecticide use in past year	Respondents	19	18			
	No	4 (21.1)	9 (50.0)			
	Yes	15 (78.9)	9 (50.0)	3.75	0.86–13.11	0.09

[a]Traffic was defined as minimal (dead end or country road), moderate (neighborhood and local roads) and heavy (main thoroughfare or highways).

29.4–357.3 ng/mg creat), but this did not reach significance in univariable analyses ($p = 0.076$; Figure 2B). Aggregate molar benzene metabolites (MUCA plus PHMA) were 0.81 nmol/mg creat (range 0.03–4.41 nmol/mg creat) in cases and 0.47 nmol/mg creat (range 0.21–2.52 nmol/mg creat) in controls ($p = 0.072$; Figure 2C).

Benzene was detected in indoor air in all but one of 37 households that returned air canisters. Air benzene concentrations were comparable between ML (median 0.26 ppb, range 0.092–1.33 ppb) and control homes (median 0.22 ppb, range 0.00–1.22; $p = 0.99$; Figure S2A). None approached the US Occupational Safety and Health Administration limits for airborne benzene of 1 ppm over 8 h [39]. Indoor air benzene, as measured in a single 8-h period, was not significantly correlated with aggregate urinary benzene metabolites across all dogs ($r = 0.17$; $p = 0.32$; Figure S2B).

Metabolites of 1,3 butadiene were detectable in all pet boxer dogs (Figure 3). The metabolite DHBM was observably but not significantly higher in cases (median 1073 ng/mg creat, range 474–4228 ng/mg creat) versus controls (median 818 ng/mg creat, range 410–1485 ng/mg creat; $p = 0.12$; Figure 3A). Urinary MHB3 concentrations were comparable between cases (median 12.65 ng/mg creat; range 4.70–49.10 ng/mg creat) and controls (median 14.25 ng/mg creat, range 3.6–992.6 ng/mg creat; $p = 0.74$; Figure 3B), as were aggregate molar 1,3-butadiene metabolites ($p = 0.46$; Figure 3C).

Trichloroethylene was not found in any drinking water samples (limit of detection, 0.18 ppb). Trichloroethylene in indoor air was below the limit of detection (0.0068 ppb) in most households (32 of 37) that returned water samples but was found at low levels in one control household (0.008 ppb) and 4 case households (0.007–0.022 ppb).

3.5 | Neighborhood Deprivation and Urbanicity

Dogs with a diagnosis of ML lived in more prosperous neighborhoods with lower national ADI percentiles (median 28.0%, range 2.0%–86.0%) compared to control dogs (median 46.5%, range 4.0%–83.0%; $p = 0.048$; Figure 4A). Consistent with owner questionnaire responses regarding neighborhood type (Table 2), dogs with ML lived in counties with greater urbanicity (i.e., lower USDA RUCC; median 1, range 1–6) compared to

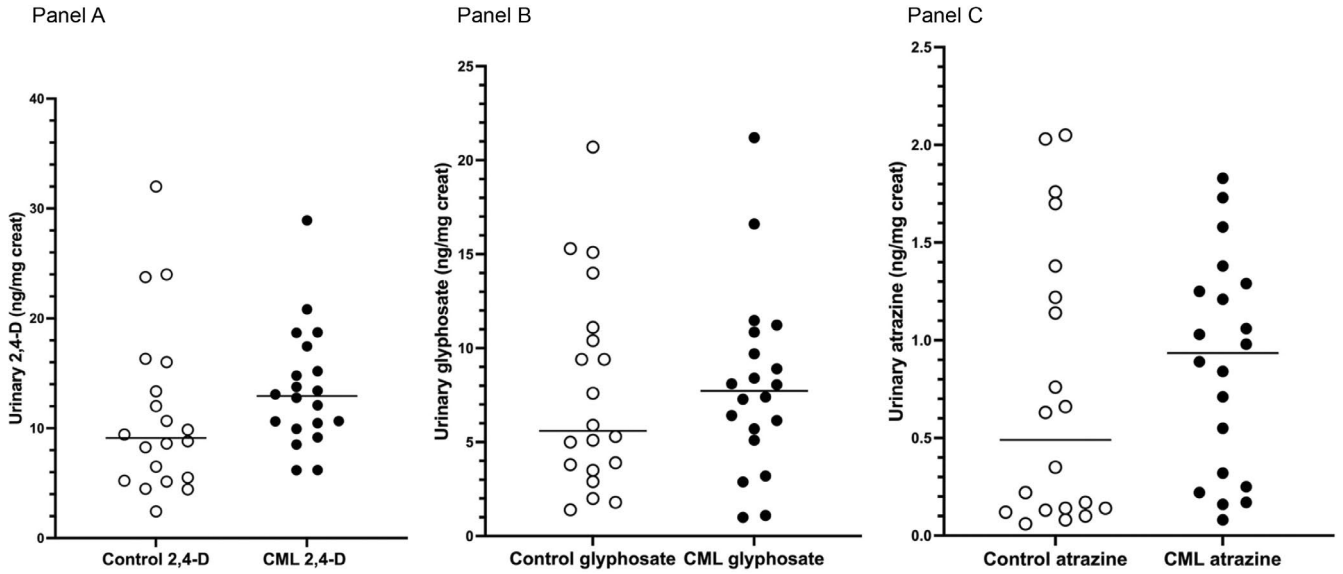

FIGURE 1 | (A) Urinary concentrations of the herbicide 2,4-D in boxers with multicentric lymphoma (ML) and unaffected controls. $p = 0.07$ between groups. Dogs were enrolled across the United States from 2021 to 2024. (B) Urinary concentrations of the herbicide glyphosate in boxers with multicentric lymphoma (ML) and unaffected controls. $p = 0.64$ between groups. Dogs were enrolled across the United States from 2021 to 2024. (C) Urinary concentrations of the herbicide atrazine in boxers with multicentric lymphoma (ML) and unaffected controls. $p = 0.29$ between groups. Dogs were enrolled across the United States from 2021 to 2024.

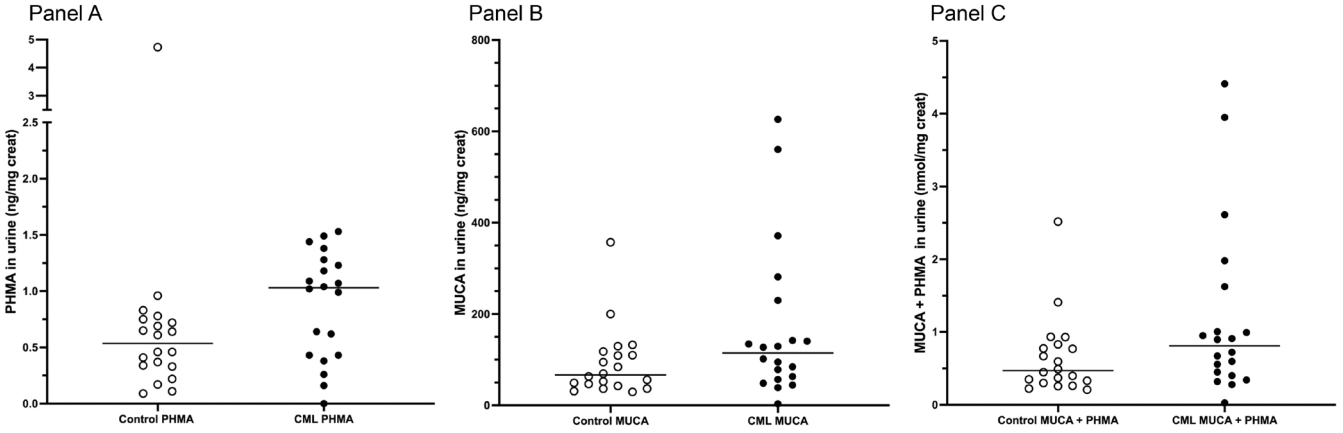

FIGURE 2 | (A) Urinary concentrations of the benzene metabolite PHMA were significantly higher in boxers with multicentric lymphoma (ML) versus unaffected controls ($p = 0.031$). Dogs were enrolled across the United States from 2021 to 2024. (B) Urinary concentrations of the benzene metabolite MUCA in boxers with multicentric lymphoma (ML) and unaffected controls. $p = 0.076$ between groups. (C) Aggregate molar urinary concentrations of measured benzene metabolites in boxers with multicentric lymphoma (ML) and unaffected controls. $p = 0.072$ between groups.

controls (median 2, range 1–8; $p = 0.032$; Figure 4B). The RUCC codes were not significantly correlated with indoor air benzene ($r = -0.04$; $p = 0.81$), aggregate urinary benzene metabolites ($r = 0.17$; $p = 0.30$) or aggregate urinary 1,3-butadiene metabolites ($r = 0.05$; $p = 0.77$).

3.6 | Multivariable Logistic Regression

Multivariable models were fitted for urinary VOC metabolite concentrations and ML outcome, evaluating ADI, RUCC, and dog body weight as covariables with backward stepwise selection. In the final models, dogs with ML had marginally but significantly increased urinary MUCA concentrations when adjusted for urbanicity (OR 1.01, 95% CI 1.00–1.02, $p = 0.035$; Table 3). Dogs with ML had non-significantly different urinary DHBM concentrations when adjusted for area deprivation (OR 1.002, 95% CI 1.000–1.004, $p = 0.065$; Table 3). Urinary PHMA was no longer significantly different between cases and controls when considering these co-variables.

4 | Discussion

All enrolled boxer dogs had detectable levels of the herbicides 2,4-D, glyphosate, and atrazine in their urine. Occupational and household exposure to these three herbicides is linked to NHL in people [11–15]. In dogs, owner-reported herbicide application is associated with lymphoma [18–20]; however, no previous studies have assessed in vivo exposures in the setting of ML.

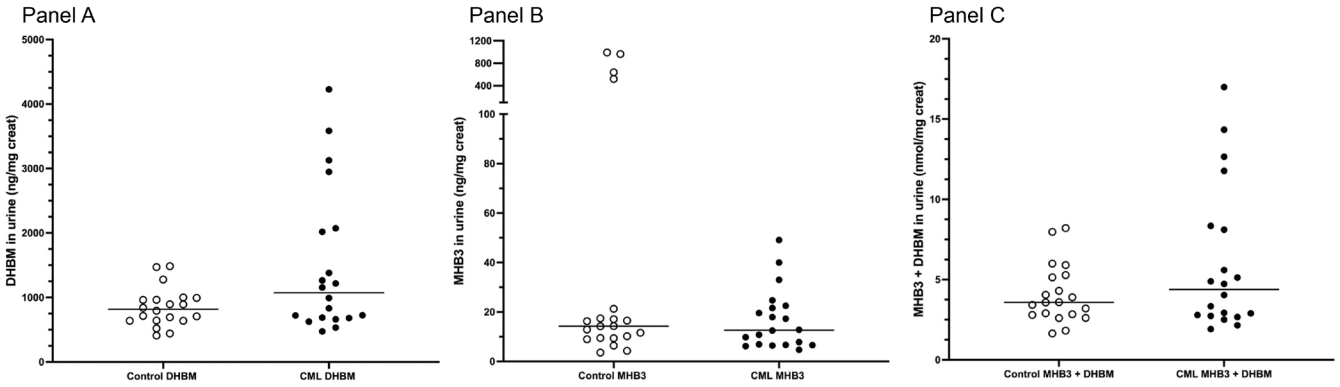

FIGURE 3 | (A) Urinary concentrations of the 1,3-butadiene metabolite DHBM in boxers with multicentric lymphoma (ML) and unaffected controls. $p = 0.12$ between groups. Dogs were enrolled across the United States from 2021 to 2024. (B) Urinary concentrations of the 1,3-butadiene metabolite MHB3 in boxers with multicentric lymphoma (ML) and unaffected controls. $p = 0.74$ between groups. (C) Aggregate urinary molar concentrations of the measured 1,3-butadiene metabolites MHB3 and DHBM in boxers with multicentric lymphoma (ML) and unaffected controls. $p = 0.46$ between groups.

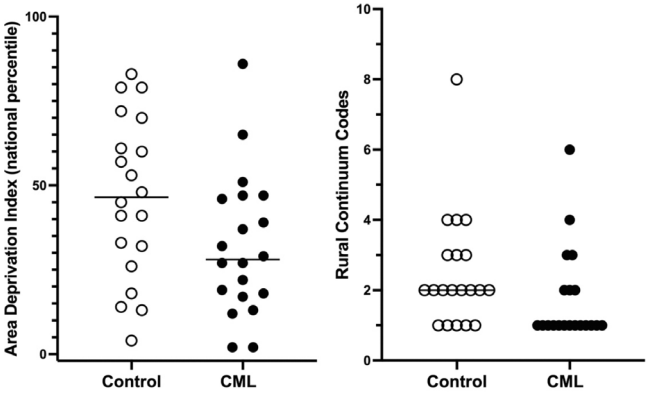

FIGURE 4 | Social deprivation and urbanicity data for boxers with multicentric lymphoma (ML) and unaffected boxer controls. (A) Area Deprivation Indices as national percentiles based on census block for each dog's home address. A higher percentile indicates greater poverty. $p = 0.048$ between cases and controls. (B) Urbanicity for each dog household based on county of home address using the USDA Rural–Urban Continuum code. A higher number indicates a more rural environment. $p = 0.032$ between cases and controls. Dogs were enrolled across the United States from 2021 to 2024.

TABLE 3 | Results of multivariable logistic regression analyses for urinary metabolites of benzene (PHMA and MUCA) and 1,3-butadiene (DHBM and MHB3) in pet boxer dogs.

Urinary metabolites	OR for exposure	95% CI	*p*
PHMA[a]	1.83	0.65-5.15	0.250
MUCA[b]	**1.01**	**1.00-1.02**	**0.035***
DHBM[a]	1.002	1.000–1.004	0.065
MHB3[a]	0.99	0.98-1.01	0.289

Note: Odds ratios (OR) are for urinary chemical exposures (in ng/mg creat) in dogs with multicentric lymphoma (ML) relative to unaffected control dogs, adjusted for the following covariates: Kg body weight, Area Deprivation Index (ADI) and urbanicity (RUCC)Dogs were enrolled across the United States from 2021 to 2024. Bolding highlights a significant difference. *$p < 0.05$.
[a]Final model adjusted for area deprivation.
[b]Final model adjusted for urbanicity.

Observed urinary 2,4-D concentrations in our boxer dog study group (2.4–32.0 ng/mg creat) were comparable to those detected in healthy control dogs in a bladder cancer case–control study (16.4–35.0 ng/mg creat) [31]. However, we observed an unexpected 13-fold variability in urinary 2,4-D in our boxer dogs. This could relate to varied timing of recent 2,4-D exposure. The urinary half-life of 2,4-D in dogs is about 4 days [40]; consistent with this, the highest urinary concentrations of 2,4-D in dogs were previously detected within 2 days of known herbicide application, with a marked decline within 16 days post-exposure [41]. Subsequent studies should consider using pooled urine samples from multiple days throughout the year for herbicide exposure analyses.

Urinary 2,4-D concentrations were observably higher in boxer dogs with ML compared to controls; however, they were not statistically different in our study group size ($p = 0.07$). Based on the large variability in observed urinary 2,4-D concentrations, a post hoc sample size calculation estimates the need for 189 dogs in each group to demonstrate this difference between ML and control dogs to be significant, with 80% power and 95% confidence, should such a difference exist.

We did not find differences in aggregate urinary herbicide concentrations by either owner reported use or seasonality of sampling. Together, these findings suggest exposures to herbicides beyond direct contact with treated grass. However, no herbicides were detected in any household water samples in the current study. Each of these herbicides is readily detected in indoor household dust [42, 43], with 2,4-D found in nearly 95% of Iowa households [44]. Glyphosate has even been detected in commercial dog foods [45], which could contribute to non-seasonal herbicide exposures. A logical next step is to survey the relationship between systemic herbicide exposures in dogs and herbicide concentrations found in indoor dust and in fed foods.

All enrolled boxers had measurable exposures to the VOC benzene. Urinary concentrations of the benzene metabolite MUCA ranged from 3 to 627 ng/mg creat across our boxer study group, which is comparable to those reported in healthy dogs of various breeds (geometric mean, ~96 ng/mg creat) [46]. Urinary PHMA

(also called SPMA) was below the limit of detection in most dogs in that same study [46]. We found urinary PHMA concentrations ranging from undetectable to 4.7 ng/mg creat in our boxers.

Median urinary concentrations of the predominant benzene metabolite MUCA did not reach statistical significance between cases and controls. After adjusting for urbanicity, dogs with ML had marginally higher urinary MUCA concentrations. Because of high individual variability in urinary MUCA concentrations, particularly among ML cases, a post hoc sample size calculation estimates the need for 292 dogs in each group to establish a significant difference in urinary MUCA concentrations in univariable analyses, with 80% power and 95% confidence; should a difference exist.

We also found that urinary concentrations of the benzene metabolite PHMA were significantly higher in ML cases compared to controls in univariable analyses. While the odds of higher PHMA concentrations in cases were not significant when adjusted for co-variables in this small study group, this initial observation in dogs is consistent with the association between NHL and benzene from both occupational and non-occupational exposures in people [6, 38] (although urinary PHMA measurements were not included in these studies). Indoor household sources of benzene include gas stoves [47], as well as tobacco smoke, attached garages, and building materials such as paints, particleboard, plywood, adhesives, and caulks [48]. Benzene is also found in household solvents and airborne crematoria emissions, both of which have been associated epidemiologically with lymphoma in dogs [22, 49]. Subsequent questionnaires should include more in-depth questions about garage location, stove type and frequency of use, and recent indoor painting, construction, or furniture assembly.

Benzene levels in indoor air did not correlate with aggregate urinary benzene metabolites across all dogs, at least when measured on a single day. We had owners collect indoor kitchen air for 8 h, and an 8-h period might have been too short or variable to correlate with urinary benzene metabolites. Most of our households were non-smoking, but we did not control for the time of day, recent use of a gas stove or cleaning products, or type of indoor ventilation. Measuring benzene in indoor air across multiple days would have been preferable, but this is cumbersome. An alternative approach to measuring environmental benzene exposures over time is to use a personal monitor such as silicone wristbands [50] adapted for dog collars [51].

We found much higher levels of the 1,3-butadiene urinary metabolites DHBM (410–4228 ng/mg creat) and MHB3 (4–993 ng/mg creat) across our study group compared to a previous study in healthy dogs of various breeds (DHBM ~115 ng/mg creat; MHB3 not detected) [46]. The previous study recruited dogs from the Albany, NY area (some from animal shelters), while our pet boxers were recruited across the country. The higher exposures in our study might reflect these differences in regional recruitment.

Despite relatively high urinary concentrations, 1,3-butadiene metabolites did not differ between boxers with ML and controls in our study. Again, we observed unexpectedly high variability in VOC metabolites, especially for DHBM within ML cases (Figure 3A). Based on our data, 51 dogs in each group would be needed to establish significantly higher urinary DHBM in ML cases, should such a difference exist. In addition, directly measuring environmental 1,3-butadiene exposures with personal silicone dog tag monitors might be informative [50, 51].

Exposures to VOCs and other environmental chemicals in people are increased in urban areas [34, 35]. In the current study, dogs with ML were more likely to live in an urban/suburban neighborhood compared to control boxers, as assessed by owner questionnaire responses and RUCC data on household neighborhood. This is consistent with the correlation between higher urbanicity and NHL in people [33] and more industrial neighborhoods and lymphoma in dogs [22]. It is difficult to tease out risk from VOC exposures compared to other exposures related to urbanicity. While urinary MUCA was still a significant risk factor for ML when controlled for urbanicity in our study, this needs confirmation in a larger sample size.

Boxers with a diagnosis of ML were from significantly more prosperous neighborhoods (with lower ADI scores) compared to control boxers, which is the opposite of what has been found for NHL [32]. We suspect that this discordance between prosperity and ML versus NHL might be due to the bias of easier access to veterinary care in more prosperous communities [52]. Regional differences in access to veterinary care (e.g., more specialty centers in urban areas) could also have biased our finding of higher urbanicity for ML. Specifically, our study could have missed dogs that developed ML in poorer rural neighborhoods without referral centers or family resources to establish a definitive diagnosis. In retrospect, we could have reduced selection bias due to differences in access to care by using population-based sampling of controls from the same veterinary clinics as cases. Studies of environmental chemical exposures and cancer risk in pets will continue to be hampered without expanded access to care and standardized veterinary cancer registries in the United States.

Our study has several other important limitations. First, our final sample size was small despite recruitment outreach nationwide. Increasing our control to case ratio could also have increased power. However, we were limited by budget, since surveying urinary and household chemicals is costly (more than $1000 per dog). We intentionally focused on a single breed, the boxer, to minimize variability in heritable contributions to ML risk. While breed-related risk factors are likely fixed, environmental risks for lymphoma in dogs are potentially modifiable. Based on our findings in boxers, further assessing benzene exposure as a risk factor for ML is an important next step in a larger group of boxers and in other breeds.

Another limitation is that our sampling at the time of recruitment only assumes that single time point measurements can be used as surrogates for chronic chemical exposures. Although the biological half-lives of VOC metabolites have not been established in dogs, urinary metabolites are much more stable than plasma VOC concentrations in people [53]. However, urinary MUCA and DHBM concentrations can still vary by about 1.9-fold across repeated sampling dates in the same individuals [54]. Subsequent studies could incorporate urinary data over several time points, along with silicone tag personal monitors [50, 51] to better assess integrated VOC exposures.

In addition, there might have been selection bias in our cases and controls, since some controls were recruited directly through owners while most cases were recruited through veterinary clinics. This could have biased the finding of higher urbanicity in cases versus controls, which in turn could have biased measured VOC exposures. Our results should be confirmed in larger prospective studies with recruitment from the same veterinary clinics.

Overall, we found a significant association between ML and both urbanicity and urinary benzene metabolites in boxer dogs. These findings are consistent with studies of NHL in people. If benzene exposures are causally related to ML as they are for NHL, this would support a recommendation that owners of dog breeds at high risk for ML consider established methods to decrease household benzene exposures. These include activated carbon indoor air filtration units [55] and even phytoremediation with indoor plants, which can decrease indoor benzene concentrations by more than 65% under controlled conditions [56, 57].

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Disclosure

Authors declare no off-label use of antimicrobials.

Ethics Statement

Approved by University of Wisconsin-Madison Institutional Animal Care and Use Committee, protocol V005450-R01. Authors declare human ethics approval was not needed.

Conflicts of Interest

Lauren A. Trepanier serves as Associate Editor for the Journal of Veterinary Internal Medicine. She was not involved in the review of this manuscript. The other authors declare no conflicts of interest.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Aggregate molar urinary

concentrations of the herbicides 2,4-D, glyphosate and atrazine in boxer dogs, by owner reported use of herbicides in the past year (Panel A, $p = 0.46$) and season of sample collection (Panel B, $p = 0.35$). Dogs were enrolled across the United States from 2021 to 2024. **Figure S2:** (A) Indoor air benzene measurements in homes of boxers with multicentric lymphoma (ML) and unaffected controls. $p = 0.99$ between groups. Dogs were enrolled across the United States from 2021 to 2024. (B) Indoor air benzene concentrations (8-h measurements in a single day) versus aggregate urinary benzene metabolites in pet boxer dogs ($r = 0.17$; $p = 0.32$). Dogs were enrolled across the United States from 2021 to 2024.